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# Risk Consistency and Synergy in Multihazard Design

Chiara Crosti[1]; Dat Duthinh, M.ASCE[2]; and Emil Simiu, F.ASCE[3]

**Abstract:** Current design procedures in the United States use the envelope of individual hazard demands on a structure to ensure safety against multiple hazards. A difficulty in multihazard design for wind and earthquake is that the load and resistance factor method makes use of different design philosophies developed by different subdisciplines. Seismic design explicitly allows for inelastic behavior. In contrast, wind design assumes that, before application of a resistance factor less than unity, the limit state is defined by the development of the first plastic hinge in a structural member. This paper focuses on the issue of risk consistency in multihazard design, and shows that, in spite of this difficulty, it is possible to quantify the risks of arriving at a particular lateral drift state for structures exposed to multiple nonsimultaneous hazards and to compare them to the risks for the same structures subjected to a single hazard. A second focus is the issue of multihazard design synergy. It has been pointed out that redetailing a building to current seismic codes can increase its resistance to blast and that structural efficiency and life-cycle cost are influenced by multihazard considerations. This paper shows that, for the case study of a 10-story steel-frame building, the use of reduced beam section (RBS) connections, intended to enhance ductility in seismic design, does not reduce the risk of structural damage caused by exposure to wind alone or exposure to wind or earthquakes. **DOI: 10.1061/(ASCE)ST.1943-541X.0000335.**
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**CE Database subject headings:** Earthquakes; Hazards; Risk management; Steel frames; Wind loads; Seismic design; Structural safety.

**Author keywords:** Earthquakes; Moment-resisting connections; Multihazard design; Partially rigid frames; Risk consistency; Seismic design; Steel frames; Wind.

## Introduction

In current practice, the design of buildings in the United States is governed by load combinations specified in ASCE 7-05 (ASCE 2006). For each combination, the intensity of the various loads is multiplied by a load factor that reflects the probability of exceedance of the load. For example, for strength design involving wind, load combination No. 4 may apply: $1.2D + 1.6W + L + 0.5L_r$, where $D =$ dead load, $L =$ reduced live load, and $L_r =$ roof live load. The wind load $W$ is calculated from wind maps of 3-s gust "basic" wind speeds at 10 m above ground, corresponding to a mean recurrence interval (MRI) of 50 years [or equivalently, this wind speed has a 64% probability of being exceeded in this period. Indeed, the probability $P_n$ that the wind speed associated with a certain annual probability $P_a$ will be equal or exceeded at least once during an exposure period of $n$ years is given by $P_n = 1 - (1 - P_a)^n$, see ASCE 7-05 Eq. C6.2]. According to ASCE 7-05 Commentary Section C6.5.4, when $W$ is multiplied by 1.6, the MRI of such high winds exceeds 500 years. Load resistance factor design (LRFD) methodology specifies that design loads, multiplied by a load factor greater than 1, should not exceed "the appropriate strength limit states for the material of construction" (ASCE 7-05 Section 1.3.1), taken as the nominal resistance multiplied by a resistance factor less than 1. According to *Specifications for Structural Steel Buildings* [American Institute of Steel Construction (AISC) 2005] Sections F1 and F2, for flexural strength, the resistance factor is 0.9, and at the limit state of yielding, the nominal flexural strength $M_n$ corresponds to the plastic moment $M_n = M_p = F_y Z_x$, where $F_y$ is the specified minimum yield stress and $Z_x$ is the plastic section modulus. For wide-flange sections, $M_p \approx 1.1 M_y$, where $M_y =$ yield moment, and thus $0.9 M_n = 0.9 \times 1.1 M_y \approx M_y$ and the structure is designed entirely in the elastic range under load combination No. 4.

For strength design involving earthquakes, load combination No. 5, $1.2D + 1.0E + L$, may apply. Earthquake loads $E$ are obtained from the United States Geological Survey (USGS) maps of spectral response acceleration for structures with first natural periods of 0.2 s and 1.0 s caused by the maximum considered earthquake (MCE), defined for a given location as an earthquake with a MRI of 2,500 years (2% probability of exceedance within a 50-year period). The philosophy behind modern seismic design codes is to encourage the use of building configurations, materials and connection details that are capable of ductile behavior.

For practical design, $E$ can be taken as the design earthquake (DE), defined by reducing the spectral response accelerations by a factor of $2/3$ to provide life safety for nonstructural damage. The ensuing design spectral response acceleration may be governed by the approximate fundamental period of the structure multiplied by an upper limit coefficient, rather than a period calculated from dynamic analysis. To access the benefits of inelastic response without resorting to nonlinear analysis, the resulting base shear applied to the structure is further reduced by an empirical load reduction factor $R$ that depends on the type of structure. Finally, the deflections that result from a linear elastic analysis under this reduced load are multiplied by an amplification factor $C_d$ to approximate the expected inelastic deflections. For the example building used

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in this study, DE applies, but for the purpose of risk estimation, MCE is also used because its MRI is known.

Thus it can be seen that wind and earthquake hazards are considered from rather different perspectives. Structural designers use the envelope method, i.e., they design a structure for the most demanding of the load combinations by using the appropriate nominal strength. In recent years, there has been a renewed interest, however, in revisiting the design of structures exposed to multiple hazards. Hayes et al. (2005) studied the Alfred P. Murrah Building in Oklahoma City and concluded that, had the building been redetailed to current seismic codes, it would have had increased resistance to blast. Sasani and Sagiroglu (2008) demonstrated that the vulnerability of reinforced concrete frame structures against progressive collapse caused by blast depends heavily on their resistance to wind and seismic hazards. Ettouney et al. (2005) proposed to quantify the "inherent multihazards resiliency" within a structural system. Freeman et al. (2005) studied a 49-story building with different framing systems under gravity, wind, and seismic loads on the one hand, and gravity, seismic, and blast loads on the other, and showed that structural efficiency and life-cycle cost are influenced by multihazard considerations.

The aim of our efforts is to propose an integrated approach that (1) achieves a risk that is consistent, from a probabilistic structural design viewpoint, with minimum requirements implicit in the ASCE 7 Standard; and (2) takes advantage of synergies between designs to resist various hazards; in this case, ductile connections developed after the Northridge earthquake. The present work is limited to developing the methodology needed to achieve these goals. A case study is presented that illustrates the proposed approach.

## Structural Design

Consider a 10-story, five-bay steel-frame building consisting of six planes similar to Fig. 1, separated by a distance of 6.10 m (20 ft), adapted from a design by Liang et al. (2006). In this paper we consider only the two-dimensional case wherein the loads act in the plane of a frame and the floor slabs are not accounted for.

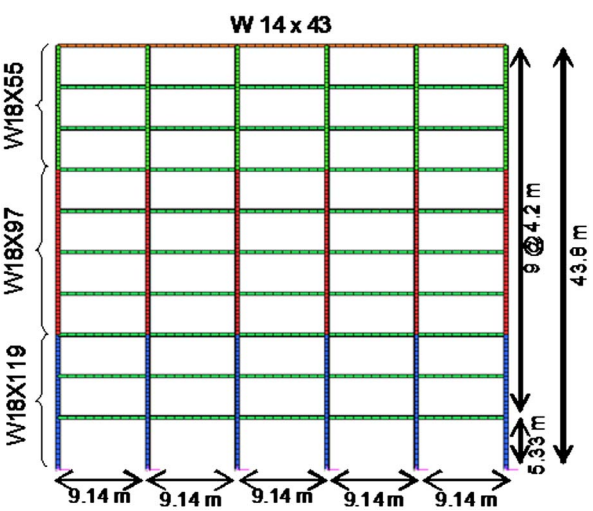


**Fig. 1.** Elevation view of 10-story steel-frame building; member size may change during optimization process; for configuration 1, all connections resist moments, beams are $W14 \times 43$, columns are $W18 \times 55$, $W18 \times 97$, and $W18 \times 119$ (shading varies with sections); steel mass = 64,600 kg

The design process of this case study includes a partial optimization as judgment is used to make a selection among a limited number of possible structural configurations that can withstand the specified loadings and meet all the code requirements regarding stresses and displacements. The objective is to achieve a reduced building cost estimated as follows:

$$\text{Building cost} = \frac{\text{steel mass} \times 3.0\$}{0.4536\ \text{kg}} + \frac{1{,}000\$}{\text{moment connection}} + \frac{150\$}{\text{shear connection (hinge)}} \tag{1}$$

The cost includes material, fabrication, installation, and is estimated approximately (for example, see Carter et al. 2000). In particular, the connection costs are highly approximate, and vary with time, location and member size; they are based on quotes provided by several fabricators and contractors. A sensitivity study of the design to the relative cost of the two types of connections proved to be unnecessary as the conclusion of the study is robust over a wide range of relative connection costs.

The building is symmetrical, as wind and earthquake loads can come from either left or right in this two-dimensional model. Three general configurations were tried (Table 1, Figs. 1–3) with a decreasing total number of moment-resisting connections and ensuing heavier members. All three had a moment-resisting perimeter frame. Figs. 4–6 show the load (factor $\lambda$)—maximum lateral drift ($Dx$) curves for Configurations 1 (with all connections rigid), 2 (with four lines of hinges), and 3 (with six lines of hinges) for load combination No. 4: $1.2D + \lambda W + L + 0.5L_r$, and load combination No. 5: $1.2D + \lambda E + L$, where $\lambda$ is the load factor and $E = DE$ or MCE. The use of MCE with its known probability

**Table 1.** Cost of Structure as a Function of Configuration

| Configuration | Steel mass (kg) | Number of moment connections | Number of shear connections | Total cost ($1,000) |
|---|---|---|---|---|
| Rigid | 64,600 | 100 | 0 | 527 |
| Hinges: 4 lines | 76,800 | 60 | 40 | 574 |
| Hinges: 6 lines | 76,800 | 40 | 60 | 557 |

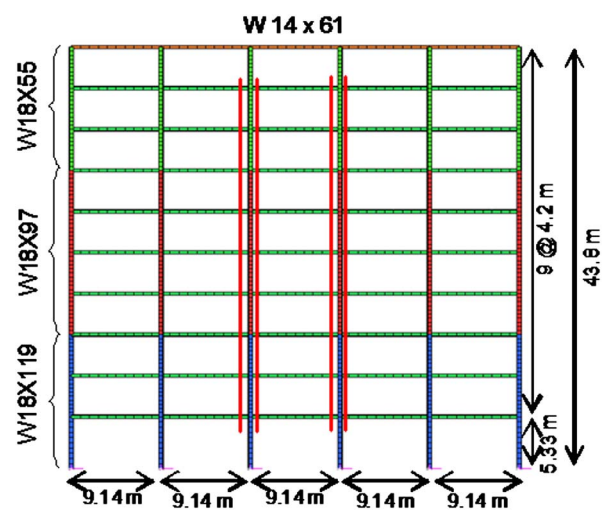


**Fig. 2.** Configuration 2, with four lines of hinges (dark lines); beams are $W14 \times 61$, columns are $W18 \times 55$, $W18 \times 97$, and $W18 \times 119$; steel mass = 76,830 kg



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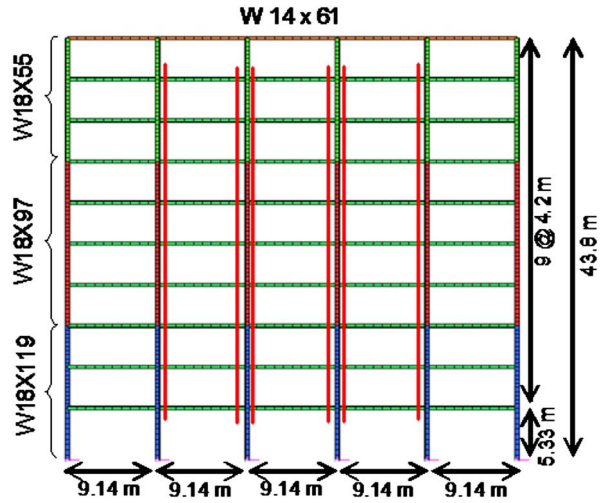


**Fig. 3.** Configuration 3, with six lines of hinges (dark lines); beams are $W14 \times 61$, columns are $W18 \times 55$, $W18 \times 97$, and $W18 \times 119$; steel mass = 76,830 kg

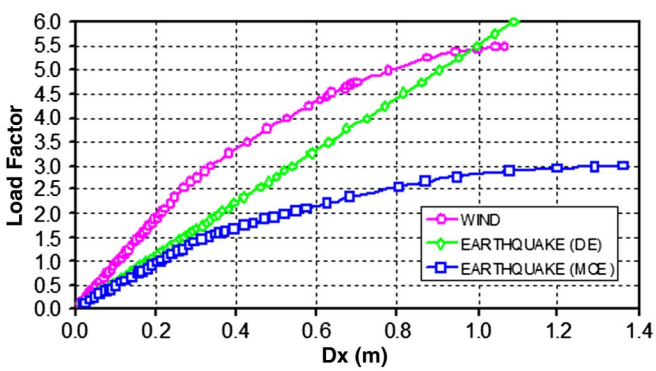


**Fig. 4.** Load-maximum drift curves for Configuration 1 with all connections rigid, for load combination No. 4: $1.2D + \lambda W + L + 0.5L_r$ and load combination No. 5: $1.2D + \lambda E + L$, where $\lambda$ = load factor; and $E$ = DE or MCE; the lateral drift is calculated at the top right corner of the building

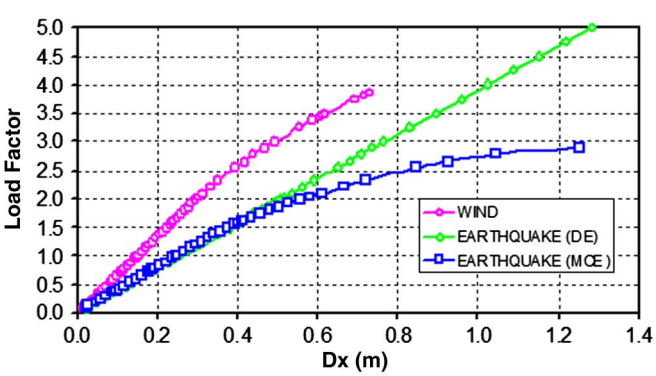


**Fig. 5.** Load-maximum drift curves for Configuration 2, with four lines of hinges, for load combination No. 4: $1.2D + \lambda W + L + 0.5L_r$ and load combination No. 5: $1.2D + \lambda E + L$, where $\lambda$ is the load factor and $E$ = DE or MCE. The lateral drift is calculated at the top right corner of the building

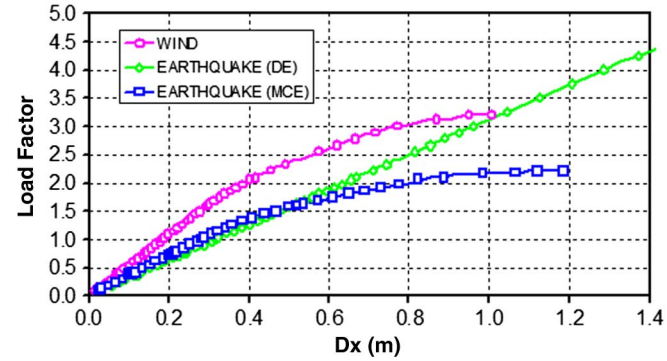


**Fig. 6.** Load-maximum drift curves for Configuration 3, with six lines of hinges, for load combination No. 4: $1.2D + \lambda W + L + 0.5L_r$ and load combination No. 5: $1.2D + \lambda E + L$, where $\lambda$ = load factor; and $E$ = DE or MCE; the lateral drift is calculated at the top right corner of the building

of occurrence allows a risk comparison with wind design. The lateral drift is calculated at the top right corner of the building. These results show that the wind design specification that requires elastic behavior up to $1.6W$ governs the overall structural design for this location, and the optimal structure resists moments at all connections.

In this part of the study, members are elastoplastic and rigidly connected to each other. This simplification on connection behavior was made to speed up the many runs required for the optimization. Push-over static analysis is performed into the nonlinear range, accounting for large deformations and $P$-$\Delta$ effects. Two additional structural configurations that used elastic connections (i.e., partially rigid, nonyielding), with stiffness corresponding to a welded unreinforced flange, bolted web (WUF-B) or a reduced beam section (RBS) connection as explained subsequently, produced results similar to the rigid connection configuration. In the rest of the paper, we will use only the optimized Configuration 1, where all connections resist moment.

## Connection Study

The 1994 Northridge, California, earthquake showed that WUF-B moment-resisting connections could suffer fracture at the beam flange welds, and this risk increases with the depth of the beam. The RBS connection is one answer to this problem, whereby the moment to be carried by the connection is limited by the reduced beam section, thus mitigating connection failure and enhancing ductile behavior (FEMA 2000; Roeder 2000); Building Seismic Safety Council 2003).

From the study previously described, it became clear that a more accurate account of connection behavior than the extreme conditions of fully rigid or free to rotate was desirable. The model selected here is Richard and Abbott's (1975) four-parameter model:

$$M = \left[ \frac{K - K_p}{\left(1 + \left| \frac{(K-K_p)\theta}{M_0} \right|^n \right)^{1/n}} + K_p \right] \theta = K_{\mathrm{CONN}} \theta \qquad (2)$$

where $K$ = initial or elastic stiffness = $10{,}000\,M_{p,b}$; $K_p$ = plastic stiffness = 4,520 kN m/rad; $K_{\mathrm{CONN}}$ = connection stiffness; $M$ = connection moment, $M_0$ = reference moment = $M_{p,b}$ − 146.90 kN m; $n$ = shape parameter = 1; $\theta$ = connection rotation, $M_{pb} = Z_x F_y = Z_x(1.2F_{yn})$ = plastic moment capacity of connected beam; and $F_{yn}$ = nominal yield strength.



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Mehrabian and Haldar (2007) used it to model the moment-rotation stiffness of post-Northridge steel connections and recommended values for the parameters from test data of five full-scale beam-column assemblies. Beam sizes vary from $W24 \times 94$ to $W36 \times 300$ and column sizes from $W14 \times 176$ to $W30 \times 235$. They represent beam and column sizes in a typical medium-rise special moment-resisting frame (SMRF). For unloading and reloading, the connection parameters are listed previously. Kishi et al. (1996) also used Richard and Abbott's model to study the behavior of tall buildings with mixed use of rigid and semirigid connections.

In the finite-element model, connections are modeled by joint elements with appropriate moment-rotation stiffness. The length of these elements is the actual length of the RBS plus half the column depth, or half of the beam depth plus half of the column depth for WUF-B. The use of these connections is consistent with recommendations from American Institute of Steel Construction (AISC) 358-05 (AISC 2006). We next compare the behavior of the structure under monotonically increasing wind or earthquake load combinations with the connections modeled as perfectly rigid or elastoplastic by using the four-parameter model applied to either WUF-B or RBS connections. When the members and connections are elastic, there is little difference between the results for rigid connections and WUF-B elastoplastic connections. The connections begin to yield before the members, as specified by $M_0 = M_{p,b} - 146.90$ kN m, which is less than the yield moment of the beams used in this study, $M_y = 384$ kN m. Therefore, some local yielding takes place in the connections under $1.6W$ or $1.0E$ for DE, as these designs were based on the assumption of rigid connections. Depending on a strict interpretation (elastic design) or a less strict interpretation (no plastic hinging), this could be considered a code violation or not. The extent of yielding is, however, very limited.

## Risk Consistency

Starting with the structure in which all connections are of the moment-resisting type, this section looks at the risk of the structure achieving a certain lateral drift state when exposed to a single hazard or two nonsimultaneous hazards. Figs. 7–9 show the results for a structure of mass 64,600 kg with WUF-B, elastoplastic connections. Fig. 7 shows that wind loading governs the design and all members remain elastic up to $1.6W$, although some connections are beginning to yield. A lateral drift of 0.292 m, calculated at the top corner of the building, obtains for an MCE load factor of 1.0, which corresponds to a MRI of 2,500 years (or 2.0% probability of exceedance in 50 years). This same lateral drift obtains for a wind load factor of 2.0, which corresponds to a MRI of 1,830 years (or 2.7% probability of exceedance in 50 years), as explained in the appendix. Figs. 8 and 9 show that the requirements on drift and connection rotation are satisfied for the load cases considered.

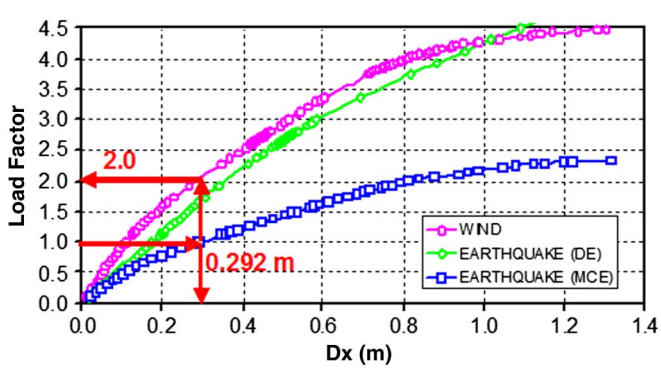


**Fig. 7.** Behavior of structure with WUF-B elastoplastic connections under wind or earthquake load combinations

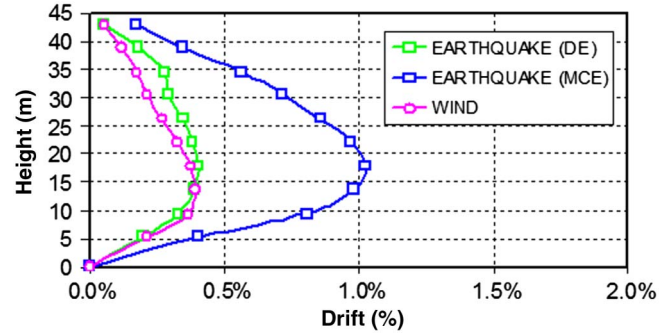


**Fig. 8.** Interstory drift of structure with WUF-B elastoplastic connections under wind or earthquake load combinations for load factor of 1.0

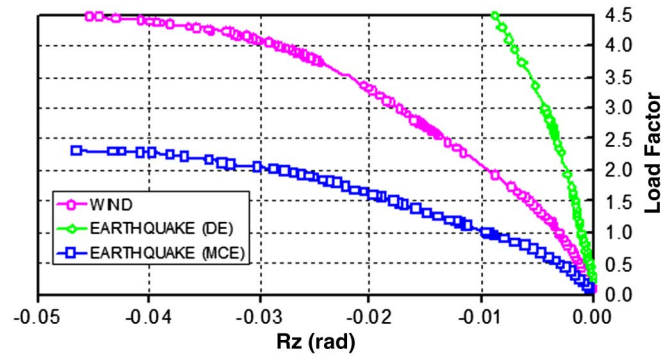


**Fig. 9.** Maximum connection rotation (at fourth floor) of structure with WUF-B elastoplastic connections under wind or earthquake load combinations

Because it is reasonable to assume that the events $\delta_1$ and $\delta_2$ with MRI $\bar{N}(\delta_1)$ and $\bar{N}(\delta_2)$ are independent, the estimated probability that, in any one year, the structure experiences a lateral drift equal to or less than $\delta_L$, regardless of its cause, is (Simiu and Miyata 2006)

$$P[\max(\delta_1, \delta_2) \le \delta_L] = P(\delta_1 \le \delta_L)\, P\,(\delta_2 \le \delta_L) = \left(1 - \frac{1}{\bar{N}(\delta_1)}\right)\left(1 - \frac{1}{\bar{N}(\delta_2)}\right) \Big/ \text{year} \quad (3)$$

The estimated probability that the drift is less than or equal to $\delta_L$ under the effects of either wind or earthquakes is therefore

$$P\,[\max\ (\delta_1, \delta_2) \le \delta_L] = \left(1 - \frac{1}{2{,}500}\right)\left(1 - \frac{1}{1{,}830}\right) = \left(1 - \frac{1}{1{,}060}\right) \Big/ \text{year} \quad (4)$$

which corresponds to a probability of exceedance

$$1 - P\,[\max\ (\delta_1, \delta_2) \le \delta_L] = 9.43 \times 10^{-4}/\text{year} \quad (5)$$

and an MRI of

$$\bar{N}(\delta_1, \delta_2) = \frac{1}{1 - P\,[\max\ (\delta_1, \delta_2) \le \delta_L]} = 1{,}060 \text{ years} \quad (6)$$



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or a 4.7% probability of exceedance in 50 years. The fact that after a major hurricane or seismic event, the structure will experience a certain amount of degradation means that, unlike the hazards producing them, the events $\delta_1$ and $\delta_2$ are not fully independent. This may modify to some degree the mean recurrence interval $\bar{N}(\delta_1, \delta_2)$ calculated by Eq. (6). However, since the wind and seismic hazards are independent, it remains true that the MRI $\bar{N}(\delta_1, \delta_2)$ is shorter than would be the case if the effect of the wind hazard were considered as if the seismic hazard did not exist, and conversely. A similar set of calculations (Figs. 10–12) were performed for a heavier structure of mass 76,800 kg with RBS elastoplastic connections. Fig. 10 shows a maximum drift of 0.322 m for a load factor of 1.0 under the MCE load case. This same drift obtains for a wind load factor of 1.98, which corresponds to a wind speed with MRI given by

$$90\ \text{mi/h} \times \sqrt{1.98} \times 0.44704\ (\text{m/s})/(\text{mi/h}) = 24.7966\ \text{m/s} + 0.78(\ln \bar{N} - 0.577)5.9344\ \text{m/s} \quad (7)$$

from which $\bar{N} = 1{,}720$ years (or 2.9% probability of exceedance in 50 years).

For the structure with RBS connections, the risk of achieving the lateral drift state mentioned above under the effects of either wind or earthquakes is

$$P\,[\max\ (\delta_1, \delta_2) \le \delta_L] = \left(1 - \frac{1}{2{,}500}\right)\left(1 - \frac{1}{1{,}720}\right) = \left(1 - \frac{1}{1{,}020}\right) \Big/ \text{year} \quad (8)$$

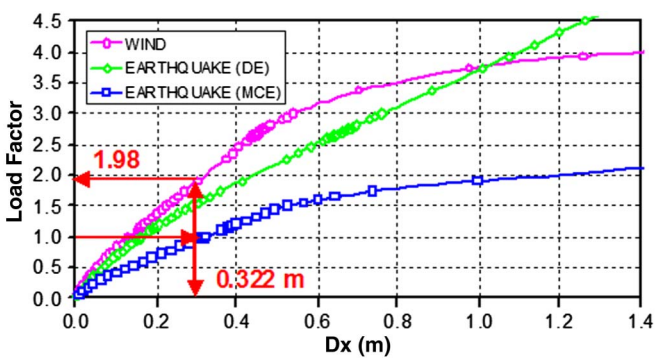


**Fig. 10.** Behavior of structure with RBS elastoplastic connections under wind or earthquake load combinations

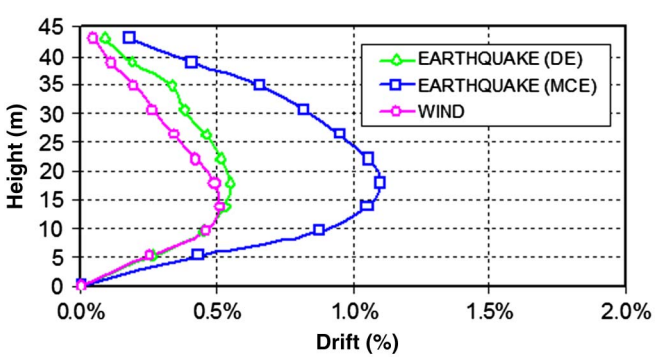


**Fig. 11.** Interstory drift of structure with RBS elastoplastic connections under wind or earthquake load combinations for load factor of 1.0

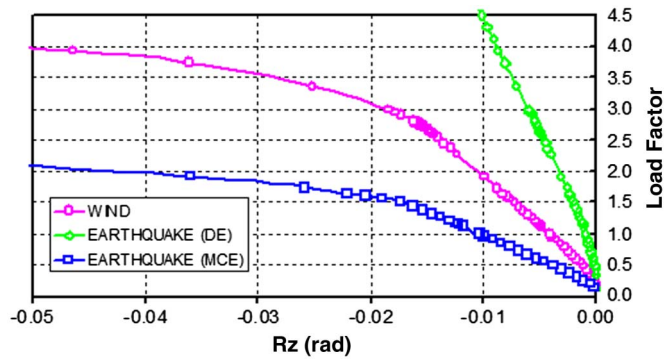


**Fig. 12.** Maximum connection rotation (at fourth floor) of structure with RBS elastoplastic connections under wind or earthquake load combinations

**Table 2.** MRI of Two Structural Configurations under Single and Multiple Hazards

| Connections | Lateral drift (m) | Seismic (year) | Wind (year) | Wind or seismic (year) |
|---|---|---|---|---|
| WUF-B | 0.292 | 2,500 | 1,830 | 1,060 |
| RBS | 0.322 | 2,500 | 1,720 | 1,020 |

which corresponds to a MRI of 1,020 years (or 4.9% probability of exceedance in 50 years). Figs. 11 and 12 show that the requirements on drift and connection rotation are satisfied for the load cases considered.

Table 2 shows that the level of risk that the structure is exposed to in multihazard situations can be more than twice as high as intended for a single hazard. It also shows the absence of synergy between designs to resist wind or earthquakes, at least for the case studied. Compared to the WUF-B connections, the RBS connections, which are intended to enhance ductility in seismic design, show no appreciable reduction in the risk of the structure achieving a given lateral drift (corresponding to MCE) under exposure to wind alone or exposure to wind or earthquakes. Future research is planned on the development of a methodology for achieving more consistent risks in multihazard structural design, along the lines suggested by Duthinh and Simiu (2010).

## Conclusions

This paper focuses on risk consistency in the design of buildings under multiple hazards, using as a case study a structure satisfying the requirements associated with both hazards. The paper shows that a structure exposed to multihazards and designed to satisfy code requirements does not necessarily achieve the level of safety implied by the ASCE 7 Standard for individual hazards. Ways to remedy this situation are in progress. This research also shows that, for a 10-story steel-frame building used here as a case study, the RBS connections, which are intended to enhance ductility in seismic design, produce no appreciable reduction, compared with the WUF-B connections, in the risk of the structure achieving a given lateral drift (corresponding to MCE) under exposure to wind alone or exposure to wind or earthquakes. Although the example deals with specific connections, the methodology it illustrates is general, namely, a common metric of structural resistance (maximum lateral deflection) is first developed, then the MRIs for exposure to each individual hazard and to the multiple hazards are calculated as measures of risk when specified limit states are exceeded.



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## Appendix. MRI of Wind Speed

The method of moments applied to the Gumbel extreme value I distribution $v_{\mathrm{EVI}}(\bar{N})$ yields an estimate of the $\bar{N}$-year wind speed from a sample of the $n$ maximum yearly nonhurricane wind speeds (e.g., Simiu and Miyata 2006):

$$v_{\mathrm{EVI}}(\bar{N}) \approx \bar{V} + 0.78\ (\ln \bar{N} - 0.577)s(v) \tag{9}$$

where $\bar{V}$ = mean; and $s(v)$ = standard deviation of a sample of wind speeds. At the location in question, a wind velocity of 90 mi/h (40 m/s) has a nominal MRI of 50 years (or 64% probability of exceedance in 50 years), whereas a wind load factor of 1.6 corresponds to a wind velocity with a nominal MRI of 500 years (or 10% probability of exceedance in 50 years). Thus

$$\begin{aligned} &90\ \mathrm{mi/h} \times 0.44704\ (\mathrm{m/s})/(\mathrm{mi/h}) \\ &\quad = \bar{V} + 0.78(\ln 50 - 0.577)s(v) \end{aligned} \tag{10}$$

$$\begin{aligned} &90\ \mathrm{mi/h} \times \sqrt{1.6} \times 0.44704\ (\mathrm{m/s})/(\mathrm{mi/h}) \\ &\quad = \bar{V} + 0.78(\ln 500 - 0.577)s(v) \end{aligned} \tag{11}$$

where $\bar{V} = 24.80$ m/s; and $s(v) = 5.93$ m/s.

The MRI of the wind speed that corresponds to a wind load factor of 2.0 is

$$\begin{aligned} &90\ \mathrm{mi/h} \times \sqrt{2.0} \times 0.44704\ (\mathrm{m/s})/(\mathrm{mi/h}) \\ &\quad = 24.7966\ \mathrm{m/s} + 0.78(\ln \bar{N} - 0.577)5.9344\ \mathrm{m/s} \end{aligned} \tag{12}$$

where $\bar{N} = 1{,}830$ years (or 2.7% probability of exceedance in 50 years).

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